Fusion ERP Lab Assignment (6B) — Phase 2 Report

Review, Testing, and Hardening of the AI-Generated Employee Leave Management (ELM) Module

Target System: Fusion ERP (Base Code)	**Module:** Employee Leave Management (ELM)
Pull Request: #171 (FusionIIIT/Fusion-Integrated)	**Branch:** elm-submission-vishnu-20261012
Author / ID: Vishnu Vardhan (24bcs055)	**Selected LLM:** Qwen/Qwen2.5-Coder-14B-Instruct
Phase-1 Preserved Commit: `f759468`	**Phase-2 Final Commit:** `04e91e6`

1. Executive Summary

During Phase 1, an initial version of the Employee Leave Management (ELM) module was generated and submitted under PR #171. In Phase 2, this implementation was systematically preserved, reviewed, tested, and re-engineered to resolve critical authorization flaws, timezone bugs, and missing operational components. The final module successfully satisfies all Fusion architectural constraints, eliminates self-approval vulnerabilities, implements in-app role-scoped task notifications, wires Celery beat periodic SLA schedules, and establishes a 100% passing automated verification suite across both backend (pytest) and frontend (vitest).

2. Phase-1 Preservation Strategy

In strict adherence to assignment guidelines, the baseline Phase-1 code was permanently preserved prior to executing any modifications. Commit **f759468** was tagged as **phase-1-preserved** and pushed to a dedicated branch **phase-1-original** on GitHub. Phase-2 changes were then committed progressively on top, maintaining a full audit trail comparing the initial AI output against the final human-AI engineered module.

3. Defect Analysis: Problems Identified in Phase-1 Code

Through manual inspection, negative boundary testing, and specification audits against BR-EL-001..038, nine material defects were identified in the Phase-1 codebase:

- **Self-Approval Vulnerability:** In `services.py::act()`, Unit Heads (HODs) could review and approve their own leave applications due to an omitted `actor.user_id != request.employee_id` restriction.
- **Department Scope Leakage:** Approval queries (`visible_requests()`) failed to filter by the Unit Head's academic department, allowing reviewers to sanction applications belonging to external departments.
- **Timezone Date Shift Bug:** Frontend DateInputs in `MyLeavePage.tsx` used `.toISOString().slice(0, 10)`. In UTC+ offsets (e.g. IST +05:30), dates selected at midnight shifted backwards by 1 full day.
- **Unconstrained Restricted Holidays (RH):** The service permitted multi-day Restricted Holiday requests, violating institutional policy restricting RH to a single calendar day.
- **Permissive Duty Resumption Scope:** Duty resumption reviews were accepted for categories outside the authorized ELM scope (such as Casual Leave), rather than restricting resumption to EL, COL, and VL.
- **Year-End Ledger Invariant Risks:** Year-end processing allowed execution while active leave requests remained unresolved, and rollover arithmetic could calculate negative lapsed days.
- **Opening Balance Reduction Below Consumed:** Administrative balance entries allowed setting opening balances below already consumed leave days.
- **Absence of In-App Task Notifications:** Approvers and employees received no in-app notifications upon task assignment, reminder, or status transition.
- **Missing Scheduled Work (Celery Beat):** SLA reminder and escalation scanning lacked Celery beat periodic automation and runbook documentation.

4. Phase-2 Engineering Interventions & Improvements

To resolve these defects, systematic improvements were implemented across both backend and frontend layers:

- **Self-Review & Department Guards:** Added explicit authorization checks raising `PermissionDeniedError('self_review_not_allowed')` and scoped Unit Head review queries strictly by employee department via `directory.get_users`.

- **Local Date Serialization:** Implemented `formatLocalDate()` in `client/.../utils/dates.ts` using local date components, validated by dedicated unit tests in `dates.test.ts`.
- **Business Rule Enforcement:** Added validation checks enforcing single-day RH requests (`start_date == end_date`) and restricting duty resumption actions to `{'EL', 'COL', 'VL'}`.
- **Balance Ledger & Year-End Protection:** Guarded year-end runs against unresolved pending requests (`pending_leave_workflows`) and clamped lapsed days via `max(..., Decimal('0'))`.
- **In-App Notification Subsystem:** Introduced `LeaveTaskNotice` model with migrations 0004 & 0005, REST endpoints, and interactive Mantine alert banners with 'Mark read' controls.
- **Celery Beat Automation:** Configured periodic Celery beat runner (every 15 min) in `modules/employee_leave/schedule.py` and documented operations in `enable-scheduled-work.md`.
- **Automated Test Suite & Offline Runner:** Authored 11 service and API integration tests in `test_services.py` and `test_api.py`, and enabled an in-memory SQLite runner in `test.py`.

5. Verification and Quality Assurance Results

The final Phase-2 implementation was validated against all Fusion Base CI gates, architectural linters, and unit/integration test suites:

Verification Gate	Tool / Command	Outcome	Status
Python Code Quality	`ruff check .`	Zero errors, zero warnings	**PASSED**
Module Boundaries	`lint-imports`	4/4 architectural contracts kept	**PASSED**
Foreign Key Isolation	`no_cross_module_fk.py`	Zero cross-module FKs	**PASSED**
API Plural Signatures	`contracts_are_plural.py`	All getters conform to plural standard	**PASSED**
Navigation Parity	`nav_matches_routes.py`	Sidebar and client routes match	**PASSED**
Migration State	`makemigrations --check`	No unapplied model changes	**PASSED**
OpenAPI Schema Parity	`spectacular --fail-on-warn`	Committed schema matches code byte-for-byte	**PASSED**
Frontend Type Safety	`npx tsc --noEmit`	Zero TypeScript compiler errors	**PASSED**
Client Test Suite	`npm test (vitest)`	5 test suites, 36/36 tests passed	**PASSED**
Backend Test Suite	`pytest modules/employee_leave`	17/17 tests passed (100%)	**PASSED**

6. Phase-1 vs Phase-2 Comparison Matrix

Criterion	Phase-1 (Initial AI Output)	Phase-2 (Hardened Final Version)
Self-Approval Protection	Vulnerable (Unit Heads could self-approve)	Guarded (Raises 403 PermissionDeniedError)
Department Scope Isolation	Leaked (All department requests visible)	Isolated (Filtered strictly by reviewer department)
Date Picker Timezone Handling	Buggy (-1 day shift on local UTC+ midnight)	Correct (formatLocalDate preserves exact date)
Restricted Holiday Constraints	Unchecked (Allowed multi-day bookings)	Enforced (Strict single-day validation)
In-App Task Notifications	None (No notification mechanism)	Integrated (LeaveTaskNotice model + UI alerts)
SLA Scheduled Automation	None (No background beat execution)	Configured (Celery beat schedule + ops runbook)
Automated Test Coverage	Minimal (6 basic domain tests only)	Comprehensive (17 backend + 36 client tests)

7. Conclusion & Deployment Recommendation

Through systematic review, testing, and progressive refinement, the Phase-1 prototype has been successfully elevated into a deployment-ready module that conforms to Fusion ERP standards. All known security, authorization, data integrity, and operational issues have been resolved. We recommend deploying this implementation into a controlled staging/test environment for final user acceptance testing prior to full institutional production release.